

**Eyes on the Task: How Difficulty and Distractor Relevance Influence Attention
In University Students**

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Abstract

The cognitive load theory and attentional control theory, though distinct psychological concepts, are connected in showing that cognitive load modulates attentional focus, further influencing the balance between top-down goal-directed attention and bottom-up stimulus-driven distraction. This study examines how cognitive load and relevance of distractors affect attentional distraction in university students performing a logical reasoning-based perceptual memory task, using eye-tracking metrics. Participants ($n = 10$) from Ashoka University completed tasks of varying difficulty, with both relevant and irrelevant distractors embedded into the task. A 2×2 within-subjects design was used to examine the main and interaction effects of cognitive load and relevance of distractors on attentional distraction. Eye-tracking metrics such as fixation count, heatmaps, and gaze plots were used to analyze attentional allocation. Results indicated a significant main effect of relevance of distractors and a significant interaction effect between difficulty level and relevance of distractors on attentional distraction. Limitations include limited ecological validity and manual data analysis, which is what future research should focus on while accounting for individual differences.

Keywords: *cognitive load, attentional control, attentional distraction, relevant and irrelevant distractors, perceptual memory, visual attention, eye tracking*

Introduction

Cognitive load theory states that people have a limited capacity to process information, and when the demand for cognitive resources exceeds this capacity, performance on tasks may decline (Sweller, 1988). Attentional control theory expands on this idea by including the role of cognitive effort in controlling attention. According to this theory, cognitive load affects attentional control, especially in cases where significant mental resources are required to complete a task. Under high cognitive load, attentional resources are more devoted to the task at hand, potentially reducing distractions (Eysenck et al., 2007). Attentional control theory also explains the interaction between top-down and bottom-up attention. Top-down control is goal-directed and is used to focus on task-relevant stimuli. Bottom-up control is stimulus-driven and leads to distraction by salient, task-irrelevant stimuli. When cognitive load is high, top-down processes dominate, enhancing task focus and reducing distraction from irrelevant stimuli as irrelevant distractors compete with the relevant, task-specific distractors (Lavie, 2005; De Fockert, 2013).

Eye tracking measures gaze duration, fixation count, and saccadic movements, which have been used in previous studies to analyse attentional processes (Holmqvist et al., 2011). The measure provides insights into how cognitive load influences visual attention, particularly in the presence of distracting stimuli.

Relevant and irrelevant distractions under cognitive load have been studied in the past; however, most studies have looked at them in isolation (Forster & Lavie, 2008). To bridge this gap, this study examines how attention is impacted when both kinds of distractions are present simultaneously and at varying task difficulty levels. This is more realistic in digital contexts where our attention is always being competed for by many distractions.

Understanding how cognitive load influences attentional control has practical implications in various fields, including education, marketing, and digital interface design (Low et al., 2011). Additionally, the integration of a physiological measure such as eye tracking provides a more objective assessment of cognitive processes than self-report measures alone. This offers empirical evidence on the neural and physiological mechanisms underlying attentional control.

Research Question

“What is the effect of task difficulty and relevance of distractors on attentional distraction in university students while performing a logical reasoning-based perceptual memory task, as measured by eye-tracking data?”

Research Variables

Independent Variable

Cognitive Load

The study operationalizes cognitive load by varying task difficulty in a logical reasoning-based perceptual memory task.

The first level was the easy (less difficult) level of the task. The first level of Raven’s Progressive Matrices was used for this (John & Raven, 2003). This level will have a low cognitive load.

The second level was the hard (more difficult) level of the task. The fifth level of Raven’s Progressive Matrices was used for this (John & Raven, 2003). This level will have a high cognitive load.

Relevance of Distractor

The distraction offered to participants was manipulated at 2 levels using different types of distractors.

The first level was relevant distractors, which are operationalised as stimuli that are components of the task and can hinder the participant's responses (Forster & Lavie, 2008). In this study, the multiple possible options for the perceptual memory task, which were similar to the correct answer (target), acted as relevant distractors.

The second level involved the presence of both relevant and irrelevant distractors. Irrelevant distractors were operationalised as stimuli that are completely irrelevant to the task and are not part of the test components (Forster & Lavie, 2008). For this study, animated images of ghosts were used as irrelevant distractors.

Dependent Variable

Attentional Distraction is the dependent variable. It was operationalised as a stimulus that diverts attention from the main task at hand (*APA Dictionary of Psychology*, 2018)

Research Hypotheses

Null Hypothesis: There will be no significant effect of cognitive load or relevance of distractor on attentional distraction.

Alternate Hypothesis I: There will be a significant main effect of Cognitive Load on Attentional Distraction.

Alternate Hypothesis II: There will be a significant main effect of the relevance of the distractor on Attentional Distraction

Alternate Hypothesis III: There will be a significant interaction effect of Cognitive Load and Relevance of Distractor on Attentional Distraction

Method

Participants

The study was conducted on 10 university students aged 17-22 years at Ashoka University, Sonipat. Participants were recruited using nomination-based convenience sampling via email. All participants were administered a consent form and a sign-up sheet for the experiment via email.

Design

This experimental study employed a 2 (Cognitive load: high and low) $\times$ 2 (Distractor Relevance: Relevant vs. Irrelevant) within-subjects factorial design. All participants were administered all the conditions. It was conducted in person, in a laboratory setting. The timing and location were notified via email in advance.

Materials

The participants were asked to solve the standardized Raven's Progressive Matrices (RPM), which is a logical reasoning-based perceptual memory task (John & Raven, 2003). The test has 5 levels with increasing levels of difficulty. The participants were administered the first and fifth levels, which were the low and high cognitive load conditions, respectively. The first five participants were administered the high cognitive load condition first, and then the low cognitive load condition was administered, and vice versa for the remaining five participants. Gaze tracking was done using the Tobii Pro Fusion eye tracker and software. The experiment was conducted in 15-minute slots.

Procedure

To prevent potential bias, participants were not fully informed about the study and its objectives. Only the task procedure was disclosed to them in the beginning (Refer to Appendix C). Once consent had been obtained, eye-tracker calibration was done. The first five participants were administered the high cognitive load condition first, and then the low cognitive load condition was administered, and vice versa for the remaining five participants. There was a break given between the two conditions during which the participants could ask any questions they had. Upon completion of the task, all the participants were debriefed about the experiment.

Data Structuring and Analysis

The number of gaze points that fell on the relevant and irrelevant distractors was counted manually by 3 researchers to avoid possible error. This data was then recorded in a spreadsheet, with each participant being assigned a row and all four conditions spread over different columns. The number of gaze points on relevant and irrelevant distractors was recorded separately. Additionally, the number of correct answers given by the participants was also recorded. All this data was recorded separately for the easy and hard conditions.

After this, all data were analysed on Jamovi (version 2.3). The data was first explored through some basic descriptive statistics (mean, median, kurtosis, etc.). Then, normality tests were conducted. All four conditions were found to pass the Shapiro-Wilk test, indicating that the assumption of normality for repeated-measures ANOVA was met. Also, the study was a 2x2 within-subjects design, it was not required to include any sphericity tests. Finally, a two-way repeated Measures ANOVA was used to test our hypothesis.

Results

Tests for Normality

We examined the assumption of normality through Shapiro-Wilk tests for all four conditions. The results were as follows: Easy Relevant in Irrelevant, $W = 0.962, p = .808$; Easy Irrelevant, $W = 0.933, p = .478$; Hard Relevant in Irrelevant, $W = 0.974, p = .923$; and Hard Irrelevant, $W = 0.966, p = .851$. For additional statistical output, refer to Appendix A.

For all conditions, the p -value was greater than 0.05. Informed by this, we can conclude that all four groups were normally distributed.

Additionally, since there were only two levels for each factor ('Relevancy' and 'Difficulty'), the assumption of sphericity was already met, and thus we did not run Tests of Sphericity.

Descriptives

Descriptive statistics were calculated for each of the four conditions: *Easy Relevant in Irrelevant*, *Easy Irrelevant*, *Hard Relevant in Irrelevant*, and *Hard Irrelevant*. Each condition included 10 participants, with 2 missing values per condition. The highest mean score was observed in the *Hard Relevant in Irrelevant* condition ($M = 34.5, SD = 11.7$), followed by *Easy Relevant in Irrelevant* ($M = 21.0, SD = 8.16$), *Easy Irrelevant* ($M = 11.4, SD = 3.37$), and *Hard Irrelevant* ($M = 5.40, SD = 1.90$). Variability was greatest in the *Hard Relevant in Irrelevant* condition (Range = 39.0) and smallest in the *Hard Irrelevant* condition (Range = 7.00).

Kurtosis values ranged from -1.14 (*Easy Irrelevant*) to 0.987 (*Hard Irrelevant*), all within acceptable limits (± 2), suggesting no severe deviations in peakedness or tail heaviness.

Hypothesis Testing - Repeated Measures ANOVA

A two-way repeated measures ANOVA was conducted to examine the effects of cognitive load (task difficulty) and distractor relevance on attentional distraction, as measured by

the number of gaze points. There was a significant main effect of distractor relevance, $F(1, 9) = 66.92, p < .001, \eta^2 = .881$, indicating that participants exhibited significantly more attentional distraction toward irrelevant distractors compared to relevant ones.

The main effect of task difficulty (cognitive load) was not statistically significant, $F(1, 9) = 2.39, p = .156, \eta^2 = .210$, suggesting that overall task difficulty did not significantly influence the total amount of distraction.

However, there was a significant interaction effect between distractor relevance and task difficulty, $F(1, 9) = 17.31, p = .002, \eta^2 = .658$. This interaction indicates that the impact of distractor relevance on attentional distraction varied depending on the task difficulty. From Table A1, we can see that the number of gaze points on irrelevant distractors is higher in the easy condition. However, the overall gaze points on all distractors are not significantly different between the hard and easy conditions.

Therefore, we reject the null hypothesis and do not reject alternative hypotheses II and III. For more statistical outputs, refer to Appendix B.

Discussion

According to the results of the study, participants were more distracted by irrelevant distractors than relevant ones, regardless of task difficulty. This finding suggests that unrelated stimuli continue to affect attention even after not being related to the dataset. There was no significant main effect of task difficulty on overall attentional distraction, which could be because cognitive load is not solely responsible for shifts in attention (Pessoa, 2009). There was a significant interaction effect between task difficulty and distractor relevance, especially in low-load conditions where participants were more prone to distraction. This finding supports the

existing research, which shows that when cognitive demands are low, more attentional resources shift to irrelevant stimuli (Lavie, 2005; De Fockert, 2013).

These primary insights can offer guidance in designing environments where focus is the central idea, such as classrooms, workspaces, or digital applications. A study conducted at Carnegie Mellon University revealed that in highly decorated kindergarten classrooms, students were more distracted than in sparsely decorated classrooms (Fisher et al., 2014). The concept also has applications in digital advertising. For example, program designers can more readily display their advertising when users are performing low-effort tasks like scrolling or reading brief articles, since they are more susceptible to irrelevant distractions like pop-ups or banner ads.

Limitations and Future Directions

Our study had limited generalizability due to a small sample size and a demographically restricted sample i.e., ten undergraduate students aged 17–22 from a single university recruited through convenience sampling. Our experiment took place in a controlled laboratory, which restricted the ecological validity of our results as we employed ghost images and GIFs as contextually irreverent distractions, which differ significantly from irrelevant distractions in the actual world, which often are emotional or context-related. We did not control for individual differences in distractibility, visual sensitivity, and working memory capacity (Núñez et al., 2015). We calculated the results by manually coding gaze points, which, while validated by numerous raters, is still prone to human error. The design relied exclusively on eye-tracking data to infer cognitive load, which does not account for any internal cognitive processes such as planning or developing strategies (Wang et al., 2025).

Future research can be improved by using multiple modalities to measure cognitive load, like GSR or pupil dilation. Expanding the sample beyond a single university and including emotionally salient and ecologically valid distractions will increase the applicability of the findings. Furthermore, in an experiment like this, looking at the involvement of working memory as an additional element can provide more understanding about how selective attention works under different situations.

Conclusion

This study, using eye tracker software, demonstrates how cognitive load may affect attentional distraction when various types of distractors are present in one's visual field. Although task difficulty did not significantly influence the total number of fixations on all distractors, it did have an effect on the number of fixations specifically on irrelevant distractors. This interaction effect tells us that even in the presence of relevant, task-directed distractors, the attentional control theory stands. These results can be used in making better online interfaces for advertising, education, and more. Future research with larger, diverse samples and more ecologically valid distractions could improve our understanding of attentional distraction and expand applications in workplace design and digital media.

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Appendix A: Descriptives and Normality Check

Table A1

Descriptives and Tests for Normality

Descriptives	Easy Relevant in Irrelevant	Easy Irrelevant	Hard Relevant in Irrelevant	Hard Irrelevant
N	10	10	10	10
Missing	2	2	2	2
Mean	21.0	11.4	34.5	5.40
95% CI mean lower bound	15.2	8.99	26.2	4.04
95% CI mean upper bound	26.8	13.8	42.8	6.76
Median	22.0	11.0	35.5	5.50
Standard deviation	8.16	3.37	11.7	1.90
Variance	66.7	11.4	136	3.60
Range	28.0	10.0	39.0	7.00
Minimum	6.00	7.00	13.0	2.00
Maximum	34.0	17.0	52.0	9.00
Kurtosis	0.173	-1.14	-0.160	0.987
Std. error kurtosis	1.33	1.33	1.33	1.33
Shapiro-Wilk W	0.962	0.933	0.974	0.966
Shapiro-Wilk p	0.808	0.478	0.923	0.851

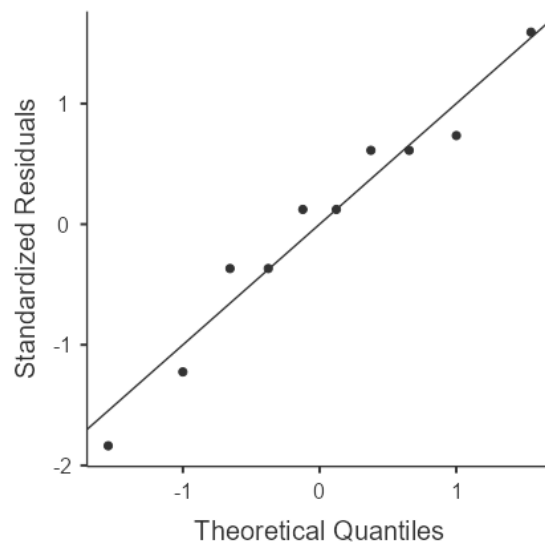
Note. The CI of the mean assumes sample means follow a t-distribution with N - 1 degrees of freedom

Figure A1

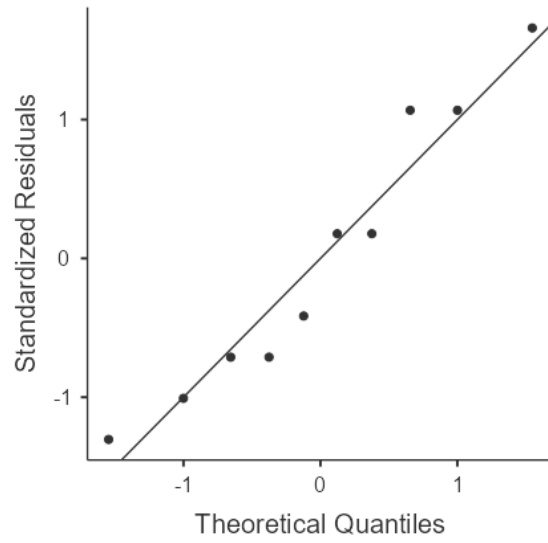
Q-Q plot of standardised residuals

A

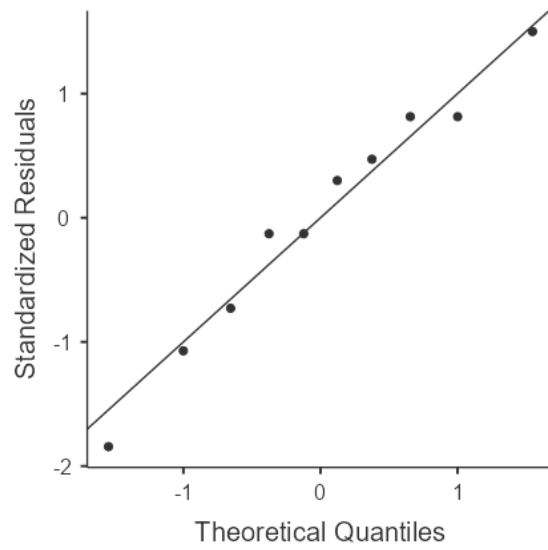
Easy Relevant in Irrelevant



B

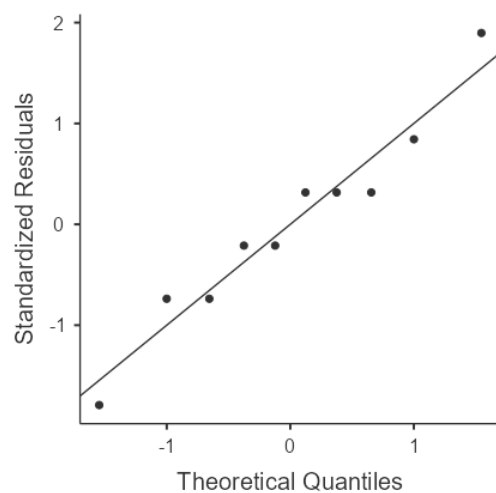
Easy Irrelevant

C

Hard Relevant in Irrelevant

D

Hard Irrelevant



Appendix B: RM ANOVA

Table B1

Repeated Measures ANOVA

Within Subjects Effects						
	Sum of Squares	df	Mean Square	F	p	η^2_p
Relevancy	3744	1	3744.2	66.92	< .001	0.881
Residual	504	9	55.9			
Difficulty	141	1	140.6	2.39	0.156	0.210
Residual	529	9	58.8			
Relevancy * Difficulty	951	1	950.6	17.31	0.002	0.658
Residual	494	9	54.9			

Note. Type 3 Sums of Squares

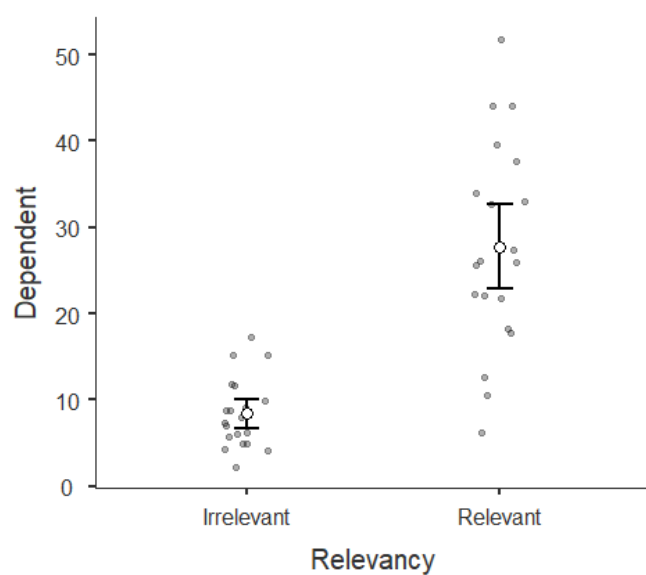
[5]

Figure B1

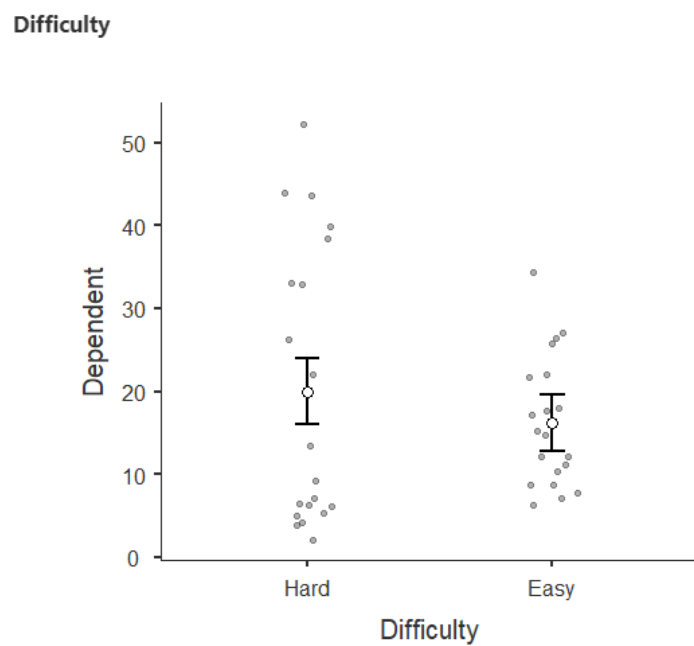
Estimated Marginal Means

A

Relevancy



B



C

Relevancy * Difficulty

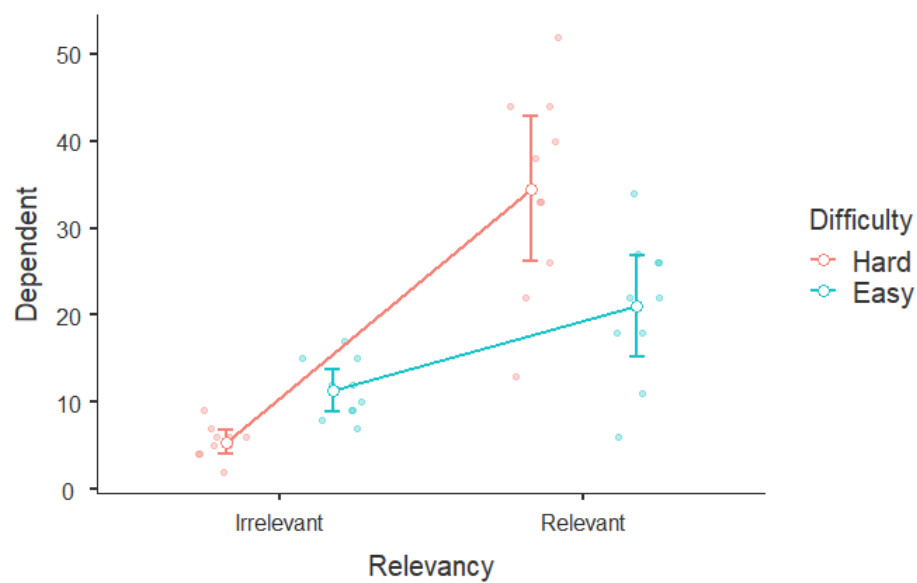


Table B2*Estimated Marginal Means*

A

Estimated Marginal Means - Relevancy

Relevancy	Mean	SE	95% Confidence Interval	
			Lower	Upper
Irrelevant	8.40	0.756	6.69	10.1
Relevant	27.75	2.152	22.88	32.6

B

Estimated Marginal Means - Difficulty

Difficulty	Mean	SE	95% Confidence Interval	
			Lower	Upper
Hard	19.9	1.75	16.0	23.9
Easy	16.2	1.51	12.8	19.6

C

Estimated Marginal Means - Relevancy * Difficulty

Difficulty	Relevancy	Mean	SE	95% Confidence Interval	
				Lower	Upper
Hard	Irrelevant	5.40	0.600	4.04	6.76
	Relevant	34.50	3.689	26.16	42.84
Easy	Irrelevant	11.40	1.067	8.99	13.81
	Relevant	21.00	2.582	15.16	26.84

Appendix C: Instructions

Instructions were given to participants at the beginning of the experiment.

Instructions for the task

- For this experiment, you will be doing a simple cognitive task. It is a pattern recognition task.
- First, you will see the question slide, which will have a pattern with some part of it missing. This slide will be there for a stipulated period and will change on its own.
- Next, you will see the answer slide with six different options. You will be using the physical mouse to click on the correct answer at your own pace. The next question will appear on your mouse click.
- There will be a black slide in the middle. If you have any questions, please ask us then.
- Since we are using an eye tracker, you will be required to keep your head still for the whole duration post calibration. The whole task will take about 10 minutes to complete.

Instructions for calibration

A dot will appear on the screen. Please follow it with your eyes as accurately as possible without moving your head.